

MACHINE LEARNING OPERATIONS

UNIT I

COMPLETE EXAM-READY NOTES

Course Code: AI254TA | RV College of Engineering
Semester V | Full Marks: 100 CIE + 100 SEE

UNIT I SYLLABUS TOPICS

What & Why | Why Now & Challenges | Defining MLOps | MLOps to Mitigate Risk | MLOps for Scale

People of MLOps: Subject Matter Experts | Data Scientists | Data Engineers | Software Engineers | DevOps | Model Risk Manager/Auditor | Machine Learning Architect

⚡ 2-MARK QUICK REVISION

Key Definitions — Memorize These!

Term	Definition (Write exactly this)
MLOps	The standardization and streamlining of machine learning life cycle management in an enterprise setting.
ModelOps	A broader term than MLOps covering any type of model (including rule-based models), not just machine learning models.
AIOps	The use of Artificial Intelligence to solve IT operational challenges (AI for DevOps). Example: Predictive maintenance for network failures.
DataOps	A practice (introduced by IBM in 2014) focused on providing business-ready data quickly with emphasis on data quality and metadata management.
Decision Automation	An increasing prevalence of decision making that happens without human intervention, powered by ML models.
Responsible AI	The responsible use of machine learning covering two dimensions: Intentionality and Accountability.
Intentionality	Ensuring that models are designed and behave in ways aligned with their purpose, including explainability and unbiased data sources.
Accountability	Centrally controlling, managing, and auditing the enterprise AI effort — no shadow IT. Ensures traceability.
Operationalization	The process of pushing ML models to production and measuring their performance against business goals.
MLOps vs Operationalization	MLOps takes operationalization further — it encompasses not just deployment but maintenance of models and the entire data pipeline in production.
Risk Assessment	Analysis based on two metrics: probability and impact of an adverse event. Severity = combination of both.
Traceability	If something goes wrong in an ML pipeline, it should be easy to find exactly where it happened.

Important One-Liners for 2-Marks

- **MLOps emerged:** The term MLOps emerged notably in late 2018 and 2019.
- **Key difference from DevOps:** Software code is relatively static but data is always changing, making ML models constantly learn and adapt.
- **Why MLOps is unique:** ML models are made up of both code AND data — not just code like traditional software.

- **Three MLOps scale goals:** Track versioning, understand if retrained models are better, ensure performance is not degrading in production.
- **AI Ops example:** Predictive maintenance for network failures — alerting DevOps teams before problems arise.
- **DataOps origin:** Introduced by IBM in 2014.
- **SMEs role:** Provide business questions/KPIs at the start and close the feedback loop at the end.
- **Data scientists' key challenge:** They are specialized in model building but not in writing applications (not software engineers).
- **Model risk two metrics:** Probability of adverse event + Impact of adverse event = Severity.



5 to 8-MARK ANSWERS

1. What is MLOps? How does it differ from DevOps? (5-8 Marks)

KEY DEFINITION: MLOps (Machine Learning Operations) is the standardization and streamlining of machine learning life cycle management. It helps organizations generate long-term value and reduce risk associated with data science, machine learning, and AI initiatives.

MLOps vs DevOps — Common Ground:

- Robust automation and trust between teams
- Collaboration and increased communication between teams
- End-to-end service life cycle (build, test, release)
- Prioritizing continuous delivery and high quality

Critical Difference:

Software code is relatively static. Data is always changing — meaning ML models are constantly learning and adapting to new inputs. This complexity (code + data together) makes MLOps a new and unique discipline.

MLOps vs ModelOps vs AIOps:

Term	What It Covers	Scope
MLOps	Machine learning model life cycle management	ML models only
ModelOps	Any type of model (ML + rule-based models)	Broader than MLOps
AIOps	Using AI to solve IT/DevOps operational problems	Separate from MLOps
DataOps	Business-ready data quality and metadata management	Data layer below MLOps

2. What are the Challenges of Managing ML Life Cycles at Scale? (5-8 Marks)

The machine learning life cycle in an enterprise setting is far more complex than it appears on the surface. There are three key challenges:

Challenge 1: Many Dependencies

- Data is constantly changing; business needs shift as well
- Results need to be continually relayed back to the business
- The model in production must align with expectations and address original goals

Challenge 2: Not Everyone Speaks the Same Language

- ML life cycle involves people from business, data science, and IT teams
- None of these groups use the same tools
- They do not share the same fundamental skills for baseline communication

Challenge 3: Data Scientists are Not Software Engineers

- Data scientists specialize in model building and assessment
- They are not experts in writing applications
- They are often forced to juggle many roles, making it challenging to do any thoroughly
- Complexity becomes exponential with staff turnover — data scientists must manage models they did not create

⚡ **EXAM TIP:** Remember: The 3 challenges are Dependencies, Language Barrier, and Data Scientists ≠ Software Engineers. These are highlighted in the textbook.

3. How does MLOps Help in Risk Mitigation? (5-8 Marks)

MLOps is important for any team that has even one model in production because continuous performance monitoring and adjusting is essential. By allowing safe and reliable operations, MLOps is key in mitigating the risks induced by the use of ML models.

Types of Risks in ML Models:

- Risk that the model is unavailable for a given period of time
- Risk that the model returns a bad prediction for a given sample
- Risk that the model accuracy or fairness decreases over time
- Risk that the skills to maintain the model (data science talent) are lost

Risk Assessment Framework:

Risk assessment is based on two metrics:

1. Probability of the adverse event
2. Impact of the adverse event

Severity = Probability × Impact (5×5 Risk Matrix)

Risks are LARGER for models deployed widely and used outside the organization.

Risk assessment should be performed at the beginning of each project and reassessed periodically.

Why Pushing Models Without MLOps is Risky:

- Full performance of an ML model can often only be assessed in the production environment
- Training data must reflect production data — if production changes, model performance drops rapidly
- ML models are sensitive to software versions and operating system environments
- Models built on open-source libraries require matching versions in production

Responsible AI — Two Dimensions:

Dimension	Meaning
Intentionality	Models designed to behave aligned with purpose. Includes unbiased data, explainability, and checks on model bias.
Accountability	Central control, management, and audit of enterprise AI — no shadow IT. Ensures data is reliable and collected per regulations. Tied to traceability.

⚡ **EXAM TIP:** Responsible AI is closely tied to MLOps — teams must have good MLOps principles to practice Responsible AI. This interplay is a common exam question.

4. Explain MLOps for Scale (5 Marks)

MLOps is an essential component to massively deploying machine learning efforts and benefiting from economies of scale. Going from one or a handful of models in production to tens, hundreds, or thousands that have a positive business impact requires MLOps discipline.

Good MLOps Practices Help Teams:

- Keep track of versioning, especially with experiments in the design phase
- Understand whether retrained models are better than previous versions (and promote better models to production)
- Ensure (at defined periods — daily, monthly, etc.) that model performance is not degrading in production

Transparency at Scale:

- Upper management and C-suite should understand what ML models are deployed in production
- They should understand the effect ML models have on the business
- MLOps provides the ability to drill down and understand the full data pipeline (from raw data to final output)



10-MARK DEEP ANSWERS

1. Explain in detail the People of MLOps and their roles (10 Marks)

MLOps is not just for data scientists. A diverse group of experts across the organization has a critical role in the ML model life cycle. The following are the key people involved:

1. Subject Matter Experts (SMEs)

- The ML model life cycle starts and ends with SMEs
- Come to the table with clearly defined goals, business questions, and/or KPIs
- Work with data scientists to frame the problem before data exploration begins
- Post-production: Close the feedback loop because traditional accuracy metrics alone are not enough
- Example: A churn prediction model with high accuracy may fail if marketing cannot prevent churn — SMEs provide this business context
- MLOps provides SMEs with model performance in business terms, not just technical metrics
- Business Decision Modeling: Helps SMEs structure and describe their needs formally

2. Data Scientists

- Build models that address business questions brought by SMEs
- Deliver operationalizable models for the production environment
- Assess model quality in tandem with SMEs
- Challenge: Technical training doesn't always prepare them for communication with business-side SMEs
- MLOps needs: Automated model packaging, visibility into all deployed models, drill-down into data pipelines, ability to deploy winning A/B test models
- Key point: MLOps must provide efficiency AND transparency to data scientists

3. Data Engineers

- Optimize the retrieval and use of data to power ML models
- Data pipelines are at the core of the ML life cycle — data engineers are at the core of data pipelines
- Work with business teams to identify the right data; work with data scientists to resolve data plumbing issues
- MLOps needs: Visibility into all deployed models + ability to drill into individual data pipelines to address underlying issues

4. Software Engineers

- Integrate ML models into the company's applications and systems
- Ensure ML models work seamlessly with other non-ML applications
- ML code, training, testing, and deployment must fit into the CI/CD pipelines
- Example: Retail site recommendation engine — software engineers integrate the ML model with the wider website

- MLOps needs: Versioning, automatic tests, ability to work in parallel on same application (e.g., Git branching)

5. DevOps

- Build and conduct operational systems; test for security, performance, and availability
- Manage CI/CD pipeline
- MLOps was born out of DevOps principles but they cannot be run as completely separate siloed systems
- MLOps need: Integration into larger DevOps strategy — systems must be complementary
- Allow automated tests for ML just like traditional software tests

6. Model Risk Manager / Auditor

- Crucial in regulated industries (especially financial services)
- Analyze model outcomes AND the initial goal/business questions to minimize overall risk
- Involved at the very beginning of the life cycle (with SMEs), post-development (pre-production compliance), and during monitoring
- MLOps need: Robust, likely automated, reporting tools on ALL models (current and historical), including data lineage
- Their work is often painstakingly manual — MLOps standardization gives a huge leg up in speed of auditing

7. Machine Learning Architect

- Ensure a scalable and flexible environment for ML model pipelines from design to production
- Introduce new technologies when appropriate to improve ML model performance
- Requires deep knowledge of BOTH enterprise architecture AND machine learning
- Collaborates across the enterprise (data scientists, engineers, DevOps, software engineers)
- MLOps need: Centralized view of resource allocation; high-level overview of models and resources consumed; ability to drill into data pipelines to assess infrastructure needs
- Role: Strategic/tactical — find long-term improvements and bottlenecks, NOT quick operational fixes

⚡ **EXAM TIP:** For 10-mark answers, write all 7 roles with their 'Role in ML Life Cycle' AND 'MLOps Requirements' — the table in the textbook is examinable!

Summary Table — All Roles at a Glance:

Role	Key Responsibility	MLOps Requirement
Subject Matter Expert	Define business goals/KPIs; evaluate model performance against business needs	Easy way to view model performance in business terms; feedback loop mechanism
Data Scientist	Build and deploy models; assess model quality	Automated packaging; visibility into all models; pipeline drill-down
Data Engineer	Optimize data retrieval and pipelines to power ML models	Visibility into deployed models; drill-down into individual pipelines

Software Engineer	Integrate ML models into applications; maintain CI/CD fit	Versioning; automatic tests; parallel development support
DevOps	Build operational systems; manage CI/CD pipelines	Seamless integration of MLOps into DevOps strategy
Model Risk Manager	Minimize ML model risk; ensure regulatory compliance	Automated reporting on all models; data lineage tracking
ML Architect	Ensure scalable, flexible ML infrastructure; introduce new tech	High-level resource overview; pipeline drill-down for infrastructure needs

2. Explain MLOps, its challenges, risks, and importance for Scale (10 Marks)

MLOps (Machine Learning Operations) is the standardization and streamlining of machine learning life cycle management. It has emerged as a critical discipline in the enterprise due to the increasing scale and complexity of deploying ML models.

Why MLOps is Needed NOW:

- Until recently, ML model numbers were manageable at small scale
- With decision automation, models have become more critical and managing model risks more important
- As more data scientists push more ML models into production, MLOps becomes critical
- Teams are looking to formalize a multi-stage, multi-discipline, multi-phase process

Three Key Challenges of ML Life Cycle at Scale:

- **Dependencies:** Data constantly changes; business needs shift; results must be relayed back continuously
- **Communication Gap:** Business, data science, and IT teams use different tools and have different skills
- **Role Mismatch:** Data scientists are model builders, not software engineers — stretched too thin at scale

MLOps vs DevOps — Critical Difference:

Software code is relatively static. ML models are made of code + data. Data is always changing, so ML models are constantly learning and adapting. This fundamental difference is why DevOps cannot be directly transferred to data science teams and why MLOps is a new discipline.

Risk Mitigation with MLOps:

- MLOps becomes critical when a team has more than a handful of operational models
- Without MLOps, having a global view of all model states is difficult
- Production environment is the ONLY place where full ML model performance can be assessed
- Risk 1: Model unavailability | Risk 2: Bad predictions | Risk 3: Decreasing accuracy | Risk 4: Loss of talent
- Risk severity = Probability × Impact (5×5 risk matrix)

MLOps and Responsible AI:

- Responsible AI = Intentionality (model behaves as designed + explainability) + Accountability (central audit trail)
- ML models lack transparency of traditional code — harder to demonstrate regulatory compliance
- Automation shifts accountability upward — from individual contributors to managers/executives
- Good MLOps principles are required to practice Responsible AI

MLOps for Scale — Key Practices:

- Track versioning of experiments during design phase
- Evaluate retrained models against previous versions before production promotion
- Periodically ensure model performance is not degrading (daily, monthly, etc.)
- Upper management should understand model performance and full data pipeline through MLOps transparency



DIAGRAMS EXPLAINED IN WORDS

Figure 1-1: Exponential Growth of MLOps (Google Trends)

What to write in exam:

The Google Trends graph shows the search interest for 'mlops' over time from January 2004 to around 2020. The interest was near zero from 2004 to approximately 2018. After 2018-2019, there was an exponential spike in interest — demonstrating that MLOps has 'seemingly skyrocketed into the data science lexicon overnight.' The term ModelOps showed parallel growth but is not shown on the same scale.

Figure 1-2: Simple Machine Learning Model Life Cycle

What to write in exam:

Figure 1-2 shows the simple (and misleadingly easy-looking) ML life cycle as a circular diagram with these steps in order:

1. Define business goal
2. Access, understand, and clean data
3. Build machine learning model
4. Deploy machine learning model
5. Iterate (loop back to step 1)

This figure is contrasted with Figure 1-3 to show that the simple view underplays the complexity of real enterprise ML life cycles.

Figure 1-3: Realistic Picture of ML Life Cycle in an Enterprise

What to write in exam:

Figure 1-3 shows four major stages in the realistic ML life cycle with the roles involved at each stage:

- Stage 1 — Business Questions: Line of business + Subject matter expert raise questions
- Stage 2 — Develop Models: Data engineers handle Data Acquisition + Data Preparation; Data scientists handle Feature Engineering + Model Training/Experimentation + Model Evaluation and Comparison
- Stage 3 — Prepare for Production: Data engineers, software engineers, ML architects handle Runtime Environment, Risk Evaluation, QA, Containerization, Elastic Scaling, CI/CD
- Stage 4 — Monitoring and Feedback Loop: DevOps handles Logging/Alerting, Input Drift Tracking, Performance Drift, Online Evaluation

Key Point: Different people with completely different skill sets, using entirely different tools, must work together — this is why MLOps is needed.

Figure 1-4: 5×5 Risk Matrix

What to write in exam:

The 5×5 Risk Matrix (also called a risk assessment table) has:

- Y-axis: Probability (Rare, Unlikely, Possible, Probable, Highly Probable)
- X-axis: Impact (Very Low, Low, Medium, High, Very High)
- Cells: Severity score (1-25) with classification: Minor (1-5), Moderate (4-12), Major (8-20), Severe (16-25)

Mitigation measures are based on the severity score. Higher probability + higher impact = higher severity = more mitigation required.



LAST-MINUTE REVISION SHEET

★ MUST-KNOW: Core Definitions

- **MLOps**: Standardization and streamlining of ML life cycle management
- **ModelOps**: Broader than MLOps — covers ALL model types (ML + rule-based)
- **AI Ops**: Using AI to solve DevOps/operational challenges (DIFFERENT from MLOps)
- **DataOps**: IBM 2014 — business-ready data quality and metadata management
- **Responsible AI**: Intentionality + Accountability
- **Risk Severity**: Probability × Impact

★ MUST-KNOW: 3 Challenges of ML Life Cycle at Scale

- Many Dependencies (data changes, business needs shift, continuous feedback needed)
- Not Everyone Speaks the Same Language (business + data science + IT = different tools/skills)
- Data Scientists are NOT Software Engineers (role overload, complexity grows at scale)

★ MUST-KNOW: 7 People of MLOps

- **SME**: Defines business goals, closes feedback loop, needs business-term performance view
- **Data Scientist**: Builds & deploys models, needs automated packaging & pipeline visibility
- **Data Engineer**: Manages data pipelines, needs model visibility & pipeline drill-down
- **Software Engineer**: Integrates ML into apps, needs versioning & auto-tests & CI/CD fit
- **DevOps**: Builds ops systems, manages CI/CD, needs MLOps integrated into DevOps
- **Model Risk Manager**: Ensures compliance & minimizes risk, needs automated reports & data lineage
- **ML Architect**: Ensures scalable infrastructure, needs centralized resource view & tech decisions

★ MUST-KNOW: 4 Risk Types in ML

- Model unavailability (downtime)
- Bad prediction for a given sample
- Model accuracy/fairness decreasing over time
- Loss of data science talent needed to maintain the model

★ MUST-KNOW: MLOps vs DevOps Critical Difference

Software code is static → DevOps works well.

ML models = code + data. Data is ALWAYS changing → ML models constantly learn and adapt.

→ This is why DevOps is NOT directly transferable to data science teams.

★ MUST-KNOW: MLOps for Scale — 3 Goals

- Track VERSIONING of experiments in design phase
- Understand if retrained models are BETTER than previous versions
- Ensure model performance is NOT DEGRADING in production (daily/monthly checks)

★ MUST-KNOW: Responsible AI — 2 Dimensions

- **Intentionality:** Models behave as designed + explainability + unbiased/compliant data
- **Accountability:** Central control/audit, no shadow IT, traceability, data reliability per regulations

☑ TOPICS COVERED

- MLOps definition and origin (2018-19)
- MLOps vs ModelOps vs AIOps vs DataOps
- 3 Challenges of ML Life Cycle at Scale
- MLOps vs DevOps (common + differences)
- Risk Mitigation and 5×5 Risk Matrix
- 4 Risk Types in ML Models
- Responsible AI — Intentionality + Accountability
- MLOps for Scale — 3 Key Practices
- All 7 People of MLOps (roles + MLOps needs)
- Business Decision Modeling for SMEs
- Operationalization vs MLOps distinction
- All 4 figures from the chapter explained

⚠ AREAS TO REVISE MORE

- The exact 5×5 risk matrix numbers (Minor, Moderate, Major, Severe scores)
- DataOps detail — when to use it vs MLOps and their overlap
- SME Business Decision Modeling details (Decision Model and Notation framework)
- Differences in MRM requirements across regulated vs unregulated industries
- Open-source library version sensitivity explanation in detail
- Specific examples for each role (from the textbook examples)

Good luck! Score Full Marks! 🎯